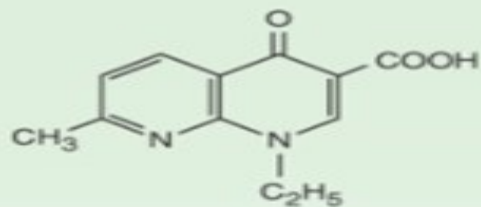


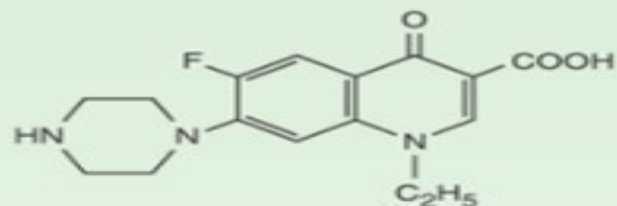
DNA gyrase inhibitors

Fluoroquinolones

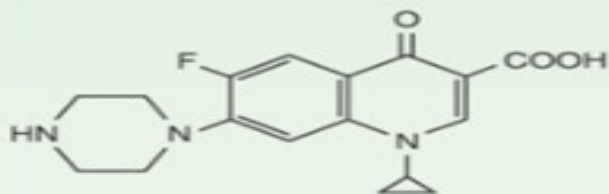
- ❖ clinically relevant quinolones are synthetic **fluorinated analogs** of **nalidixic acid**
- ❖ They are active against a variety of Gram-positive and Gram-negative bacteria.
 - nalidixic acid....was used only for UTI
- ❖ They are also active against agents of
 - atypical pneumonia**
 - Intracellular pathogens such as Legionella and
 - some mycobacteria, including MTB and Mycobacterium avium complex.



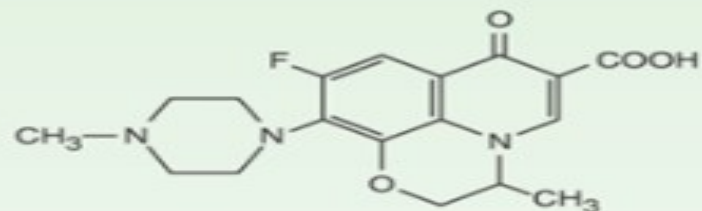
Nalidixic acid



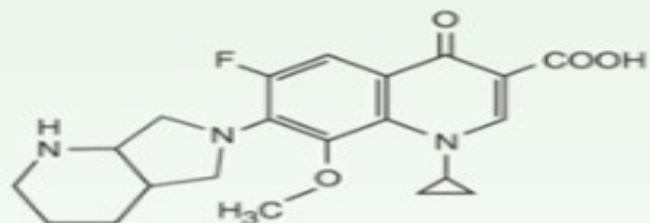
Norfloxacin



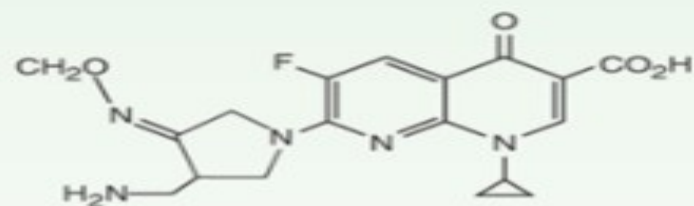
Ciprofloxacin



Levofloxacin



Moxifloxacin



Gemifloxacin

Mechanism of Action

- ❖ Quinolones block bacterial DNA synthesis by inhibiting **bacterial topoisomerase II** (DNA gyrase) and **topoisomerase IV**.
- ❖ DNA gyrase.....helps relaxation of positively supercoiled DNA that is required for normal **transcription** and replication.
- ❖ topoisomerase IV.....for **separation of replicated** chromosomal DNA into the respective daughter cells during cell division.

Table: Pharmacokinetic properties of some fluoroquinolones.

Drug	Half-Life (h)	Oral Bioavailability (%)	Peak Serum Concentration (mcg/mL)	Oral Dose (mg)	Primary Route of Excretion
Ciprofloxacin	3-5	70	2.4	500 twice daily	Renal
Gemifloxacin	8	70	1.6	320 once daily	Renal and nonrenal
Levofloxacin	5-7	95	5.7	500 once daily	Renal
Moxifloxacin	9-10	>85	3.1	400 once daily	Nonrenal
Norfloxacin	3.5-5	80	1.5	400 twice daily	Renal
Ofloxacin	5-7	95	2.9	400 twice daily	Renal

- ❖ Norfloxacin.....is the least active of the fluoroquinolones against both Gram (+) and(-), MICs 4to 8 fold higher than those of ciprofloxacin.

Ciprofloxacin

Enoxacin,

Lomefloxacin

levofloxacin,

ofloxacin, and

pefloxacin

excellent G(-) and **moderate to good** G(+)

- ❖ MSSA are susceptible but not MRSA

- ✓ In staphylococcal infections.....in combination with rifampin or other.....to **prevent resistance**

- ❖ **Enterococci** less susceptible than staphylococci,
- ❖ **Ciprofloxacin** is the most active agent against Gram-(-)
particularly **P aeruginosa**.
- ❖ **Levofloxacin**, has superior activity against Gram (+)
 - Especially **Streptococcus pneumoniae**.

Gatifloxacin, gemifloxacin, and moxifloxacin

- ❖ Have improved activity against Gram (+), particularly **S pneumoniae** and **some staphylococci**
- ❖ G (-) Activity < **Ciprofloxacin**
- ❖ Have less **MICs** for staphylococci compared to ciprofloxacin group
- ❖ Moxifloxacin has modest activity against **anaerobic** but not against **P aeruginosa**.
- ❖ Moxifloxacin is Mostly used as **ophthalmic** preparation

Resistance

resistance is due to

- ✓ one or more point mutations in the quinolone binding region
- ✓ to a change in the permeability of the organism..
- ✓ Resistance to one fluoroquinolone...cross-resistance to all other members of this class.

Pharmacokinetics

- ❖ The relatively long half-lives of levofloxacin, gemifloxacin, and moxifloxacin
 - Dose..... once-daily dosing.
- ❖ absorption is impaired by divalent and trivalent cations,
- ❖ be given 2 hours **before** or 4 **hours** after these products
- ❖ Most fluoroquinolones eliminated by renal mechanisms
- ❖ Dosage adjustment for $\text{CrCL} < 50 \text{ mL/min}$,
 - Except **moxifloxacin** and **pefloxacin** since...metabolized in the liver
 - Be given with caution in patients with hepatic failure

Clinical Uses

- ❖ Except moxifloxacin used for UTI
 - Efficacy in UTI (fluro >>TMP-SMX)
- ❖ Diarrhea
- ❖ infections of soft tissues, bones, and joints
intra-abdominal and respiratory tract
infections(except **Norfloxacin**)
- ❖ **Ciprofloxacin a choice** for prophylaxis and treatment of anthrax;
- ❖ levofloxacin is also approved by FDA for prophylaxis.

Clinical Uses...

- ❖ Azithromycin + gemifloxacin
 - for the treatment of **gonococcal infection** an alternate to ceftriaxone
- ❖ Levofloxacin and ofloxacin
 - as alternative options for **chlamydial urethritis** or **cervicitis**.
- ❖ Ciprofloxacin, levofloxacin, or moxifloxacin
 - part of a treatment regimen for TB and Non-TB
 - for eradication of **meningococci from carriers** and for
 - prophylaxis in neutropenic cancer patients.

Clinical Uses...

Respiratory fluoroquinolones

- for treatment of lower respiratory tract infections.

Adverse effects

- ❖ generally well tolerated.
- ❖ The most common effects are..... nausea, vomiting, and diarrhea.
- ❖ gatifloxacin, levofloxacin, gemifloxacin, and moxifloxacin
 - Prolongation of the QTc interval
- ❖ **Gatifloxacin** hyperglycemia in DM patients and can also causes hypoglycemic in patients receiving oral hypoglycemic agents
- ❖ Quinolones can **damage growing cartilage** and **Tendinitis**
- ❖ Neuropathy